Supplementary Materials
TAQS: Task-Aware Query-Anchored Data Selection for Scientific Machine Learning

Mingxing Jiang
Shanghai Research Institute for Intelligent Autonomous Systems, Tongji University
Shanghai, China
jiangmingxing97@163.com

Xiang Yin
Shanghai Research Institute for Intelligent Autonomous Systems, Tongji University
College of Environmental Science and Engineering, Tongji University
Shanghai Innovation Institute
Shanghai, China
glowing@tongji.edu.cn

Yuqiong Chen
College of Environmental Science and Engineering, Tongji University
Shanghai, China
2431365@tongji.edu.cn

Sijie Lin✉
Shanghai Research Institute for Intelligent Autonomous Systems, Tongji University
College of Environmental Science and Engineering, Tongji University
Shanghai Innovation Institute
Shanghai, China
lin.sijie@tongji.edu.cn

A IMPLEMENTATION DETAILS

Data preprocessing. All datasets used in this study were preprocessed to ensure compatibility with the experimental framework. Numerical variables were standardized, whereas categorical variables were encoded accordingly. One-hot encoding was applied for ED, MD, RR, and MLP.

For the ATRIV dataset, each Simplified Molecular Input Line Entry System (SMILES) string was parsed into a molecular object using RDKit cheminformatics toolkit. Compounds were represented as fixed-length numerical vectors formed by concatenating Morgan fingerprints with a set of fundamental molecular descriptors. Toxicity endpoints were then integrated into a tabular matrix with one row per compound and one column per variable.

For the DRGD dataset, CIELAB color-space illumination normalization using Contrast Limited Adaptive Histogram Equalization (CLAHE) and retinal-mask extraction were first performed. Lesion-related features (microaneurysms, hemorrhages, and exudates), vascular features (Frangi-based vessel segmentation, vessel density, fractal dimension, branching points, and tortuosity), as well as texture and color statistics (Gray-Level Co-occurrence Matrix (GLCM) and Laplacian features) were extracted from the green channel. In addition, a 512-dimensional deep embedding was obtained using a pretrained ResNet18 model and subsequently reduced to 50 dimensions via Principal Component Analysis (PCA). All extracted features were combined with labels to construct the final feature table.

Task-aware weights. To obtain stable and unbiased estimates of feature importance, global SHAP values were computed from XGBoost models. XGBoost models were optimized on 90% of each dataset through a two-stage grid search combined with 5-fold cross-validation, using R^2 as the evaluation metric. A final model was then trained on the full dataset, and the relative contribution of each feature was estimated from its mean absolute SHAP value. The resulting importance proportions remained fixed throughout the experiments and were used solely as feature weights in the similarity metric.

Hyperparameter tuning for MLP. Multi-layer perceptron regression (MLPRegressor with ReLU activation and Adam optimization) was evaluated on all six datasets using a discrete hyperparameter grid. Numerical variables were standardized and categorical variables were one-hot encoded before being provided to the network. Each candidate configuration was assessed using 5-fold cross-validation on the corresponding dataset. For each dataset, the best-performing configuration was selected and subsequently fixed for all MLP-related experiments.

Implementation. For all TAWG-related experiments, performance was evaluated using 10-fold evaluation protocol across the five mixed-type datasets.

For TAQS-related experiments under the transduction setting, a 50-fold evaluation protocol protocol was adopted. Specifically, in each fold, 98% of the data served as the candidate training set, while the remaining 2% was treated as a simulated task set. This procedure was repeated fifty times. The primary objective was to achieve unbiased coverage of the entire dataset while enabling a detailed analysis of neighborhood-size effects over the range $k \in [0, 200]$. Compared with conventional 10-fold evaluation, this strategy substantially expands the effective exploration range of k and provides a more reliable assessment of performance trends. For experiments involving stochastic components, each configuration was repeated three times using random seeds 42, 123, and 999.

Reported results correspond to the mean performance, and standard deviations are provided as measures of uncertainty.

For TAQS-related experiments under the inductive setting, 50-fold cross-validation was conducted within the candidate set to select the optimal neighborhood size k. We select k from a near-optimal performance region and favor candidates supported by neighboring k values to reduce sensitivity to isolated cross-validation peaks. The inductive setting was constructed as follows. First, the original dataset was partitioned into 20 clusters using classical Gower distance and K-Medoids based solely on feature data, without using labels. To balance local target-domain structure with task-domain shift, we selected the largest cluster containing no more than 20% of the original dataset and randomly sampled approximately 5% of the original dataset from it to form the target pool. The remaining samples from this source cluster, together with all samples from the other clusters, formed the candidate set C. The cluster-size cap was intended to prevent an overly large source cluster from reducing the mismatch between the target domain and the candidate pool, while the sampling proportion kept the target pool relatively small. The target pool was then randomly divided into disjoint task and evaluation sets, Q and E, each representing approximately 2.5% of the original dataset. This design kept the evaluation-set size comparable to that in the transductive setting. Although Q and E originated from the same target pool, they contained no overlapping samples. TAQS used only the features in Q for subset selection, whereas E was reserved exclusively for final performance evaluation. All other experimental details, including preprocessing, feature weights, and model configurations, followed the transductive setting.

Runtime and scalability experiment. For each dataset, we fixed the input split at 90% for the candidate set and 10% for the task set, with the neighborhood size set to k=50. The K-Medoids subset size was matched to that selected by TAQS. Runtime measurements covered only the data-selection stage, excluding downstream model training and evaluation. After one untimed warm-up run, each method was executed serially five times. We report the median wall-clock time and median peak memory to compare their computational efficiency and resource requirements at practical data scales.

Robustness to noisy data selection. Data corruption is introduced at both the feature and sample levels, covering continuous variables, categorical variables, and labels. Three corruption intensities (small, medium, and large) are considered, with noise injected into 50%, 70%, and 90% of the samples, respectively. We inject multi-type noise, including Gaussian perturbation, outlier corruption, categorical mislabeling, category structure distortion (merge/split), encoding drift, and composite sample noise, to simulate realistic data corruption scenarios.

Environment. All experiments were implemented in Python, performed on a computer with a 12th Gen Intel(R) Core(TM) i7-12650H 2.30 GHz CPU having 10 cores and 16 GB DDR4 main memory with SSD.

B ASSESSMENT OF POTENTIAL INFORMATION LEAKAGE

Although the optimal model parameters used for feature importance estimation were derived from an independent subset of each dataset, and the resulting feature importance weights remained fixed throughout all experiments, we nevertheless conducted a rigorous information leakage analysis. The objective was to quantify any potential performance inflation arising from the inclusion of a small number of samples that may have been involved in feature-weight estimation and later appeared in the training folds of cross-validation. Therefore, we conducted leakage analyses on representative datasets spanning different data scales.

Three representative datasets (IHDP, MHR, and OBES) were selected for evaluation. These datasets are characterized by relatively small to medium sample sizes, a higher susceptibility to overfitting, and substantial performance gains under TAQS, thereby providing a stringent testbed for leakage assessment. The analysis was conducted from two complementary perspectives: (1) weight stability, which evaluates the sensitivity of SHAP-derived weights to different data subsets; and (2) leak impact, which quantifies the influence of potential leakage on downstream predictive performance.

Weight stability (full vs. subset). To evaluate the stability of SHAP-based feature importance, we compare feature weights computed on the full dataset with those derived from subsampled data. Step 1: Full-data weights. Compute feature importance weights ω_{full} using the entire dataset. Step 2: Subsampled weights. Repeat the following procedure k times (e.g., $k = 10$): randomly sample 80% of the data, train the model, and compute SHAP values to obtain ω_{sub}^{k}. Step 3: Stability metrics. For each ω_{sub}^{k} and ω_{full}, we compute: Spearman rank correlation: measures consistency in feature ranking; L2 distance (after normalization): captures numerical differences in weights; Top-5 feature overlap: evaluates agreement on the most important features. Higher correlation, smaller L2 distance, and larger Top-5 overlap indicate greater stability.

Leak impact assessment (Cases A, B, and C). To quantify the effect of potential information leakage, we compared the performance of TAWG-based KNN regression under three weight-estimation strategies using 10-fold cross-validation. The objective was to quantify the extent to which information associated with the test folds could artificially inflate predictive performance. Case A (potential leakage): a fixed weight vector computed from the full dataset and reused across all cross-validation folds. Case B (strict no-leakage): feature weights recomputed within each fold using only the training partition, with the test fold completely excluded from weight estimation. Case C (intermediate setting): feature weights computed from an independent subset containing 80% of the samples and fixed across all cross-validation folds. The performance gap between different cases was used to quantify leakage: $\Delta_{leak} = Pref(A/C) - Pref(B/C)$, where differences in R^2 were used to assess both the existence and magnitude of potential leakage effects.

As reported in Table B.1, we report the L2 distance after normalization, where smaller values indicate higher numerical consistency. High overlap indicates strong consistency in identifying important features. Across all datasets, Spearman rank correlations exceeded

Table B.1: Feature weight stability metrics under subsampling.

Dataset	Spearman ↑	L2 ↓	TOP-5 Overlap ↑
IHDP	0.97 ± 0.02	0.02 ± 0.01	0.94 ± 0.09
MHR	0.98 ± 0.01	0.00 ± 0.00	1.00 ± 0.00
OBES	0.96 ± 0.02	0.03 ± 0.01	0.90 ± 0.10

Table B.2: Impact of information leakage on model performance: Cases A (full-data weights), B (fold-wise weights), and C (subset weights).

Dataset	A (R^2± std)	B (R^2± std)	C (R^2± std)	A-B	A-C	C-B	TAWG - VG (R^2)	TAWG vs. VG (%)	Leakage ratio ≤ 2%
IHDP	0.700±0.041	0.698±0.042	0.703±0.043	0.002	-0.003	0.005	0.315	74%	✓
MHR	0.747±0.016	0.749±0.016	0.748±0.017	-0.002	-0.001	-0.001	0.332	75%	✓
OBES	0.939±0.029	0.939±0.026	0.945±0.027	0.000	-0.006	0.006	0.261	39%	✓

0.95, indicating an extremely high degree of agreement in feature rankings. L2 distances were consistently below 0.05, suggesting that the weight vectors were numerically very similar. In addition, Top-5 feature overlap exceeded the threshold of 0.9, indicating only minimal variation in the most important features across different subsets. Overall, feature importance derived from SHAP is highly stable across different data subsets, indicating robustness to data perturbations.

Table B.2 shows that no significant performance inflation was observed when using full data feature importance. The performance gap between the potentially leakage-prone setting (Case A) and the strictly leakage-free setting (Case B) is within 2%, indicating negligible leakage effects. In contrast, the proposed method achieves an overall improvement of approximately 39–75% over the baseline. This suggests that the vast majority of performance gains (>95%) stem from the method itself rather than information leakage.

Taken together, these results demonstrate that SHAP-based feature weights are largely insensitive to data subsampling. More importantly, using feature weights estimated from the full dataset does not introduce measurable performance bias or artificially inflated results. Therefore, throughout this study, fixed feature-importance weights were used to emulate expert-prior knowledge in the similarity metric.